

Making up infant formula at a Target concentration

From an Australian or New Zealand formula label to a recipe of whole scoops and water to the nearest 5 mL.

Key terms

Concentration	Energy density of prepared formula, always in kcal per 30 mL .
Target volume / Target concentration	The bottle you want to make.
Reconstitution statement	The label's "powder + water makes this much" line, e.g. "1 scoop + 50 mL water makes approximately 55 mL".
kcal per scoop	Energy in one level scoop of powder.
Displacement	The volume (mL) one gram of powder adds beyond its water, in mL per g.
Recipe	A whole number of scoops plus water rounded to 5 mL.
Actual volume / Actual concentration	What the recipe really makes, after rounding.

Part A · Reading the label

Find: **energy per 100 mL** of prepared formula, the **Reconstitution statement**, and **grams per scoop**.

- A1** Scoops in the statement = grams of powder ÷ grams per scoop (*skip if it counts scoops*)
- A2** Displacement = (prepared volume – water) ÷ grams of powder (*3 dp; scoops × grams per scoop if it counts scoops*). Usually **0.55–0.90 mL per g** however big the scoop; if not, re-read the label.
- A3** kcal per scoop = energy per 100 mL × (prepared volume ÷ scoops) ÷ 100 (*1 dp*)

Part B · Building the recipe

- 1 kcal/mL = Target concentration ÷ 30
- 2 kcal needed = Target volume × kcal/mL
- 3 Scoops = kcal needed ÷ kcal per scoop, **rounded up** to a whole scoop
- 4 Actual kcal = scoops × kcal per scoop
- 5 Total powder = scoops × grams per scoop
- 6 Total displacement = total powder × displacement
- 7 Water = Actual kcal ÷ kcal/mL – total displacement, **rounded up to the nearest 5 mL**
- 8 Actual volume = water + total displacement
- 9 Actual concentration = Actual kcal ÷ Actual volume × 30

Check: if the Actual concentration is **0.5 kcal/30 mL or more below** the Target concentration, double-check your inputs and working before using the recipe.

Rounding

- Scoops: always round **up**, unless the answer is already exactly whole.
- Water: round **up** to a 5 mL mark, the smallest amount you can measure reliably on a bottle, unless it's already on one.
- Show kcal, grams and volumes to 1 dp, displacement to 3 dp, Concentration to 2 dp. Keep the unrounded numbers in your calculator between steps.
- So the recipe always comes out at or just **below** the Target concentration, and at or just **above** the Target volume.

Worked example: NAN OPTIPRO 1, 150 mL at 24 kcal/30 mL

NAN OPTIPRO 1: the label says

Energy: 67 kcal per 100 mL

Reconstitution statement: 129 g powder + 900 mL water makes 1 L

Grams per scoop: Average scoop weight = 4.3 g

Part A • Reading the label

	Step	Working	Answer
A1	Scoops in the statement	$129 \div 4.3$	30 scoops
A2	Displacement	$(1000 - 900) \div 129$	0.775 mL per g
A3	kcal per scoop	$67 \times (1000 \div 30) \div 100$	22.3 kcal

Part B • Building the recipe

	Step	Working	Answer
1	kcal/mL	$24 \div 30$	0.8
2	kcal needed	150×0.8	120 kcal
3	Scoops	$120 \div 22.3 = 5.38$, round up	6 scoops
4	Actual kcal	6×22.3	133.8 kcal
5	Total powder	6×4.3	25.8 g
6	Total displacement	25.8×0.775	20.0 mL
7	Water	$133.8 \div 0.8 - 20.0 = 147.3$, round up to 5 mL	150 mL
8	Actual volume	$150 + 20.0$	170.0 mL
9	Actual concentration	$133.8 \div 170.0 \times 30$	23.61 kcal/30 mL

Recipe: 6 scoops + 150 mL water

Makes: 170.0 mL at 23.61 kcal/30 mL

0.5 kcal/30 mL or more under target? No

Common mistakes

- Making the water equal to the Target volume. The powder takes up room too: subtract the total displacement.
- Rounding scoops to the nearest whole scoop instead of up.
- Forgetting to round the water up to a 5 mL mark.
- Dividing by scoops instead of grams of powder when working out displacement.
- Dividing by grams per scoop when the Reconstitution statement already counts scoops.
- Mixing up per 100 mL (on the label) with per 30 mL (Concentration).
- Leaving out the check in step 9 when the bottle is small and the scoop is large.